Total No. of Questions : 8]

SEAT No. :

P3246

[Total No. of Pages : 2

[5461]-284

B.E. (I.T.)

SOFT COMPUTING

(2012 Pattern) (Semester - I) (Elective - I) (414456A)

Time : 2½ Hours] ***[Max. Marks : 70***

Instructions to the candidates:

1) ***Figures to the right indicate full marks.***
2) ***Assume suitable data, if necessary.***

Q1) a) Comment on tpye of problems solved with soft computing. **[6]**

b) List and explain performance issues in EBP algorithm. **[8]**

c) Explain resonance in ART networks. **[6]**

OR

Q2) a) Comment on types of solutions obtained with soft computing. **[6]**

b) List out the strength and weaknesses of EBP algorithm. **[8]**

c) Explain the steps involved in KNN algorithm for clustering. **[6]**

Q3) a) What is meant by fuzzy logic? Illustrate it with proper example. **[8]**

b) Explain the Alpha-cut method for discrete fuzzy sets to perform arithmetic operations. **[8]**

i) Addition

ii) Division

OR

Q4) a) List out the characteristic features of fuzzy logic. **[8]**

b) List and explain following fuzzy set operations with example. **[8]**

i) Normal fuzzy set

ii) Product of fuzzy set

P.T.O.

Q5) a) Differentiate between evolutionary strategy and evolutionary programming? **[8]**

b) Explain operations in simple genetic algorithms. **[8]**

OR

Q6) a) Explain how genetic algorithms are different from evolutionary programming. **[8]**

b) Explain operations in genetic programming. **[8]**

Q7) a) Describe application of soft computing in semantic web. **[9]**

b) Describe application of fuzzy logic for character recognitions. **[9]**

OR

Q8) a) Describe how soft computing can be used in information retrieval. **[9]**

b) Describe how evolutionary computing is used in image processing. **[9]**

♦ ♦ ♦